

Frequency Shift Keying (FSK) and Amplitude Shift Keying (ASK) technology that had higher energy consumption. To enhance the sensitivity and achieve better performance, both the modules come with metal covering that protects them against EMI interference and offers considerably better performance, even when surrounded by noise from other devices.

Reyax

<https://reyax.com>

Camera module

e-con Systems' See3CAM_50CUGM is an all-in-one camera based on Sony's Pregius IMXX264 sensor. The camera



features a large pixel size of $3.45\mu\text{m}$ that enables it to capture

higher details in pictures, even in low light environments. It offers a high frame rate that helps to capture video data of up to 2000fps with custom resolution. The camera offers superior sensitivity and quantum efficiency (Peak QE) with its back-illuminated pixel technology and fast shutter speed, which further improves the low-light imaging of the camera and enables it to capture distortion-free images of fast-moving targets. This camera features the essential elements necessary to meet the demanding needs of the life science industry. It is suitable for use in a wide variety of areas, including medical, life sciences, and industrial applications.

e-con Systems

<https://www.e-consystems.com>

Media accelerator card

The Alveo MA35D media accelerator from AMD is a 5nm ASIC based media accelerator card designed for video acceleration for data centres. It can improve the quality of video streaming services and ultimately end users' experience while reducing the cost of operations. The media accelerator features dual ASIC based video processing units and can deliver high

channel density and power efficiency while reducing the cost of processing.

It also features an AI processor that evaluates content, frame-by-frame, and dynamically adjusts encoder settings to improve perceived visual quality while minimising bitrate. The MA35D platform provides ultra-low latency support for the mainstream H.264 and H.265 codecs and features next-generation AV1 transcoder engines delivering up to a 52% reduction in bitrate for bandwidth savings versus a comparable software implementation. It is suitable for video streaming applications, such as watch parties, live shopping, online auctions, and social streaming.

AMD

<https://www.amd.com/en/accelerators>

RT FPGA development kit

The RT PolarFire development kit is an integrated development platform that combines Microchip's flight-ready RT PolarFire FPGA with the development kit and interfaces. The development platform is suitable for evaluating design concepts based on actual



in-flight electrical and mechanical characteristics.

The RT PolarFire FPGA increases mission-critical computing and connectivity throughput compared to SRAM FPGAs while consuming up to 50% less power and offering greater immunity to radiation-induced configuration single event upsets. This platform enables designers to solve difficult spaceflight challenges like reducing satellite signal processing congestion. The development kit employs the same FPGA board that will be used in the spacecraft; therefore, it will enable designers to prototype with the same low-power, high-throughput radiation-tolerant (RT) FPGA, thus enabling them to do proper testing of all the parameters. RT PolarFire FPGA comes with MIL-

STD-883 Class B qualification, making microelectronic devices suitable for use within military and aerospace electronic systems, including basic environmental tests to determine resistance to deleterious effects of natural elements and conditions surrounding military and space operations.

Microchip

<https://www.microchip.com>

SoC for long-range IoT

The FG25 SoC from Silicon Labs is one of the most secure long-range smart city solutions with a large internal memory. It features a powerful Arm Cortex-M33 processor. FG25 can operate with minimal data loss at a range of up to 1.6km in dense urban and rural environments. It is also the first SoC in Silicon Labs' portfolio to support the orthogonal frequency-division multiplexing (OFDM) modulations introduced in Wi-SUN Field Area Network (FAN) 1.1, enabling high data bandwidth of up to 3.6Mbps. The SoC can help address the growing need for intelligent, connected devices in cities and other municipal environments. The SoC boasts advanced security features that ensure network security. Its higher and improved data rates will allow city planners to connect more smart devices to enable data-sharing and optimisation between city services, improving efficiency and saving cost.

Silicon Labs

<https://www.silabs.com>

Scalable microcontroller portfolio

Texas Instruments MSPM0 MCU portfolio is a scalable Arm Cortex-M0 + microcontroller (MCU) portfolio that



features a wide range of computing, pinout, memory, and integrated analogue options.

The MCU portfolio is supported by intuitive software and design tools. It includes MCUs running at frequency